

META-LANGUAGES

$$\langle \text{nonemptyletterseq} \rangle ::= \langle \text{letter} \rangle \mid \langle \text{letter} \rangle \langle \text{nonemptyletterseq} \rangle$$

Ex.2: Specify, using BNF, both signed and unsigned integers, with the following constraints:

- $$\langle \text{zero} \rangle ::= 0$$

$$\langle \text{nonzero_digit} \rangle ::= 1 \mid 2 \mid \dots \mid 9$$

$\langle \text{sign} \rangle ::= + \mid -$

$$\langle \text{digit} \rangle ::= \langle \text{zero} \rangle \mid \langle \text{nonzero_digit} \rangle$$
$$\langle \text{digit_seq} \rangle ::= \langle \text{digit} \rangle \mid \langle \text{digit} \rangle \langle \text{digit_seq} \rangle$$
$$\langle \text{abs_val} \rangle ::= \langle \text{nonzero_digit} \rangle \mid \langle \text{nonzero_digit} \rangle \langle \text{digit_seq} \rangle$$
$$\langle \text{int} \rangle ::= \langle \text{zero} \rangle \mid \langle \text{abs_val} \rangle \mid \langle \text{sign} \rangle \langle \text{abs_val} \rangle$$
$$C1: \begin{matrix} 12 \\ -0 \end{matrix}, \begin{matrix} 12 \\ +0 \end{matrix}, \begin{matrix} -12 \\ -0 \end{matrix}, 0$$

```
int = ["+" | "-"]not_zero{digit} | "0"
```

```
not_zero = "1" | ... | "9"
```

```
digit = not_zero | "0"
```

Sample mini-language spec

id = letter { letter | digit }
const int a = 4;
 ?

"abc" 'a' X ' ' ' \$' " "

Seminar 2. **SCANNING**

input: source.txt
tokens file
output: PIF + ST +
lex err (if any)

Program Internal Form =
array of pairs

PIF

symbol table

```
VAR    a: integer;
       b: integer;
       c: integer;

BEGIN
  a := 10;
  b := a + 10;
  WRITE("A message: ");
  WRITE(b);

END.
```

ST

only identifiers
and constants

token	ST_pos
VAR	-1
id	0
:	-1
integer	-1
;	-1
id	1
:	-1
integer	-1
;	-1
id	2
:	-1
string	-1
BEGIN	-1
id	0
:=	-1

ST_pos	symbol
0	a
1	b
2	c
3	10
4	"..."

constant	3
;	-1
id	1
:=	-1
id	0
+	-1
constant	3
;	-1
WRITE	-1
(	-1
constant	4
)	-1
;	-1
WRITE	-1
(	-1
id	1
)	-1
;	-1
END	-1
.	-1

Lexical errors

- 1) 1a → "wrong id"
- 2) # → "illegal alphabet char"
- 3) "A msg" } not enough
- 4) ", 'a, 'ab' } quotes

b := 2 ??? → this is a string not a char

Seminar 3 **GRAMMARS**

$$(ab)^2 = abab$$

$$a^2 b^2 = aab b$$

$$(ab)^2 \neq a^2 b^2$$

1. Given the grammar $G = \{N, \Sigma, P, S\}$
 $N = \{S, C\}$, $\Sigma = \{a, b\}$

$$P: S \rightarrow ab^1 | aCSb^2$$

$$C \rightarrow S^3 | bSb^4$$

$$CS \rightarrow b^5$$

a) prove that $w = ab(ab^2)^2 \in L(G)$

$$S \xRightarrow{2} aCSb \xRightarrow{4} abSbSb \xRightarrow{1} abab bSb \xRightarrow{1} abab babb \Rightarrow$$

$$\Rightarrow ab(ab^2)^2 = w$$

b) $w = a^2 bab^2$ $w \in C(G)$?

$$S \xRightarrow{2} aCSb \xRightarrow{3} aSSb \xRightarrow{1} aabSb \xRightarrow{1} aababb \Rightarrow a^2 bab^2 = w$$

2. $L = \{a^{2m}bc; m \geq 0\}$, $N = \{S\}$, $\Sigma = \{a, b, c\}$

$$P: S \rightarrow a^2 S | bc$$

$$S \Rightarrow a^2 S \Rightarrow a^2 bc$$

$$S \Rightarrow a^2 S \Rightarrow a^2 a^2 S \Rightarrow a^2 a^2 bc$$

$$S \Rightarrow a^2 S \Rightarrow a^2 a^2 S \Rightarrow a^2 a^2 a^2 S \Rightarrow a^2 a^2 a^2 bc$$

$$S \Rightarrow bc$$

$$? L = L(G)$$

1. $? L \subseteq L(G)$

$$\forall m \in \mathbb{N}, a^{2^m}bc \in L(G)$$

$$P(m): a^{2^m}bc \in L(G)$$

? $P(m)$ true, $\forall m \in \mathbb{N}$ math induction

= equal
→ production
⇒ derivation

I. verification step

$$P(0): a^0bc = bc \in L(G)$$

$S \Rightarrow bc \Rightarrow P(0)$ is true

II. proof step

Let's suppose for a given $k \in \mathbb{N}^*$ $P(k)$ is true. We will try to prove that $P(k+1)$ is also true.

$$P(k) = \text{True} \Rightarrow a^{2^k}bc \in L(G) \Rightarrow S \xRightarrow{*} a^{2^k}bc$$

$$P(k+1) = a^{2^{k+1}}bc$$

$$S \xRightarrow{1} a^2S \xRightarrow{\text{ind}} a^2a^{2^k}bc = a^{2^{k+1}}bc = P(k+1) \in L(G)$$

So from I and II $\Rightarrow P(m)$ is true, $\forall m \in \mathbb{N}$

2. $L(G) \subseteq L$

$$S \Rightarrow bc$$

$$\Rightarrow a^2S \Rightarrow a^2bc$$

$$\Rightarrow a^4S \Rightarrow a^4bc$$

$$\Rightarrow a^6S \dots$$

$$\Rightarrow L(G) \subseteq L$$

We can notice that using all productions in all possible combinations we only get as sequences of terminals sequences of shape $a^{2^m}bc$, $\forall m \in \mathbb{N} \Rightarrow L(G) \subseteq L$

$$3. L = \{0^m 1^n 2^m \mid m, n \in \mathbb{N}^*\}, G = (N, \Sigma, P, S)$$

$$N = \{S, R, P\}$$

$$\Sigma = \{0, 1, 2\}$$

$$P: S \rightarrow RP$$

$$R \rightarrow 0R1 \mid 01$$

$$P \rightarrow 2P \mid 2$$

$$S: S$$

$$? L = L(G)$$

$$(01)^m \neq 0^m 1^m$$

$$1. L \subseteq L(G), \forall m, n \in \mathbb{N}^*$$

$$0^m 1^n 2^m \in L(G)$$

$$\text{Let } m, n \in \mathbb{N}^* \text{ given}$$

$$S \xRightarrow{1} RP \xRightarrow{a) \quad m} 0^m 1^n P \xRightarrow{b) \quad m} 0^m 1^n 2^m \in L(G)$$

$$\left. \begin{array}{l} a) R \xRightarrow{m} 0^n 1^n, \forall n \in \mathbb{N}^* \\ b) P \xRightarrow{m} 2^m, \forall m \in \mathbb{N}^* \end{array} \right\} \text{ needs proof}$$

$$2. ? L(G) \subseteq L$$

$$a) S \rightarrow RP \text{ is the only production of } S$$

$$b) \text{ Tree for } R \Rightarrow R \text{ can only generate } 0^m 1^n, n \in \mathbb{N}$$

$$c) \text{ Tree for } P \Rightarrow P \text{ can only generate } 2^m, m \in \mathbb{N}$$

$$a) + b) + c) \implies L(G) \subseteq L$$

after proof

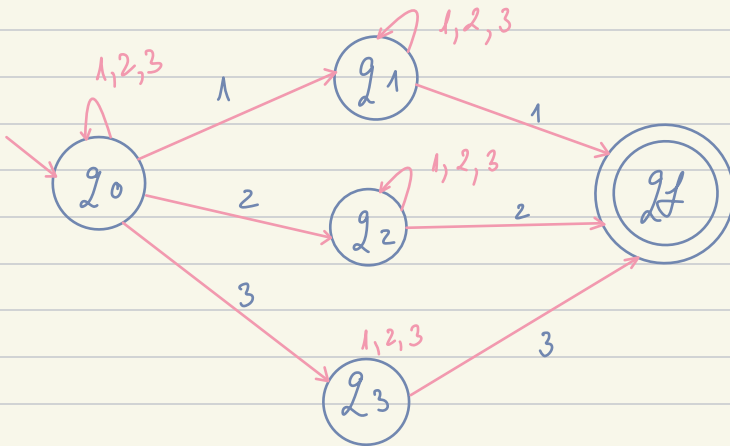
Seminar 4.

FINITE AUTOMATA (FA)

1. Given the FA: $M = (Q, \Sigma, \delta, q_0, F)$, $Q = \{q_0, q_1, q_2, q_3, q_f\}$, $\Sigma = \{1, 2, 3\}$, $F = \{q_f\}$

Σ	1	2	3
q_0	$\{q_0, q_1\}$	$\{q_0, q_2\}$	$\{q_0, q_3\}$
q_1	$\{q_1, q_f\}$	$\{q_1\}$	$\{q_1\}$
q_2	$\{q_2\}$	$\{q_2, q_f\}$	$\{q_2\}$
q_3	$\{q_3\}$	$\{q_3\}$	$\{q_3, q_f\}$
q_f	$\emptyset$	$\emptyset$	$\emptyset$

Prove that $w = 12321 \in L(M)$



$$(q_0, 12321) \vdash (q_1, 2321) \stackrel{3}{\vdash} (q_1, 1) \vdash (q_f, \varepsilon) \Rightarrow$$

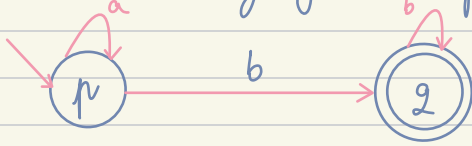
$$\Rightarrow (q_0, w) \stackrel{*}{\vdash} (q_f, \varepsilon) \Rightarrow w \in L(M)$$

i.c

f.c

initial
final

2. Find the language accepted by the FA below:



$$L = \{ a^n b^m \mid n \in \mathbb{N}, m \in \mathbb{N}^* \}$$

$$? L = L(M)$$

$$1) L \subseteq L(M)$$

$$2) L(M) \subseteq L$$

$$1) L \subseteq L(M)$$

$$\forall n \in \mathbb{N}, m \in \mathbb{N}^*, a^n b^m \in L(M)$$

Let n, m - fixed, $n \in \mathbb{N}, m \in \mathbb{N}^*$

$$\text{I) } (p, a^n) \xrightarrow{n} (p, \varepsilon), \forall n \in \mathbb{N}$$

$$\text{II) } (q, b^k) \xrightarrow{k} (q, \varepsilon), \forall k \in \mathbb{N}$$

$$(p, a^n b^m) \xrightarrow{\text{I)} n} (p, b^m) \xrightarrow{(II) m-1} (q, b^{m-1}) \xrightarrow{(II) 1} (q, \varepsilon) \Rightarrow$$

$$\Rightarrow a^n b^m \in L(M)$$

$$\text{I) } P(n): (p, a^n) \xrightarrow{n} (p, \varepsilon), n \in \mathbb{N}$$

1) Verification step:

$$P(0): (p, a^0) \xrightarrow{0} (p, \varepsilon) \text{ true}$$

2) Proof step:

$$P(k): (p, a^k) \xrightarrow{k} (p, \varepsilon), \text{ we assume true}$$

$$P(k+1): (p, a^{k+1}) \xrightarrow{k+1} (p, \varepsilon)$$

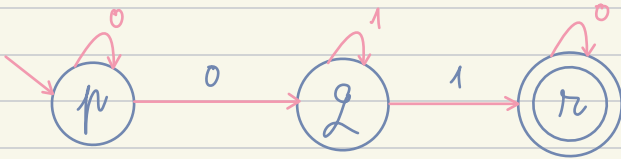
$$(p, a^{k+1}) \xrightarrow{\quad} (p, a^k) \xrightarrow[k]{P(k)} (p, \varepsilon) \Rightarrow (p, a^{k+1}) \xrightarrow{k+1} (p, \varepsilon)$$

$\Rightarrow P(k+1)$ true

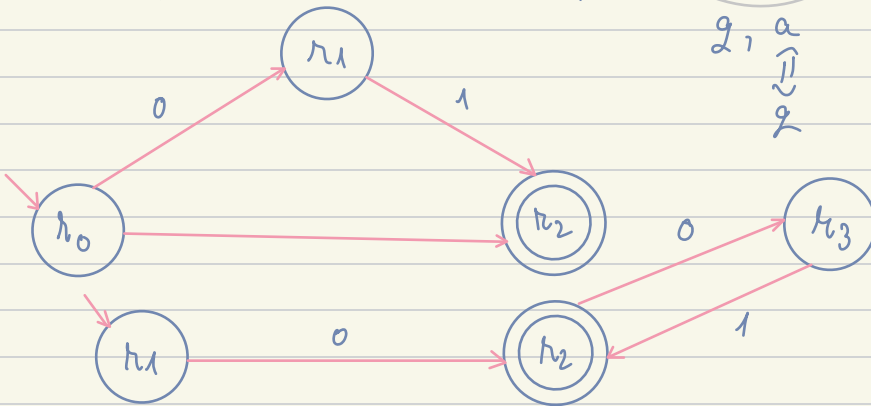
$$\begin{array}{c} a^m b b^m \\ \updownarrow \\ a^m b^m \end{array}, m, m \in \mathbb{N}$$

$$\begin{array}{c} a^m b^m \\ \updownarrow \\ a^m b^m \end{array}, m \in \mathbb{N}, m \in \mathbb{N}^*$$

3. a) $L_1 = \{0^m 1^m 0^g \mid m, m \in \mathbb{N}^*, g \in \mathbb{N}\}$



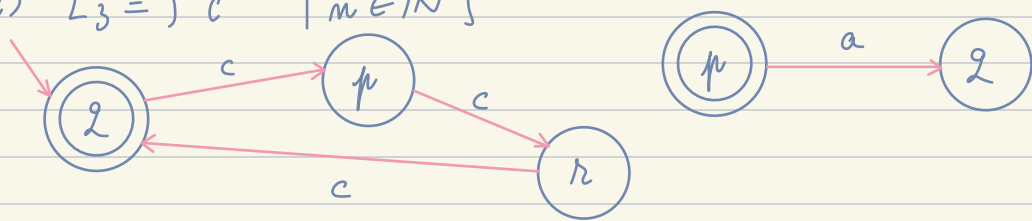
b) $L_2 = \{0(01)^n \mid n \in \mathbb{N}\}$, $f: \underbrace{Q \times \Sigma}_{\substack{q, a \\ \updownarrow \\ q}} \rightarrow \mathcal{P}(Q)$



$\varepsilon \notin \Sigma; \varepsilon \in \Sigma^*$



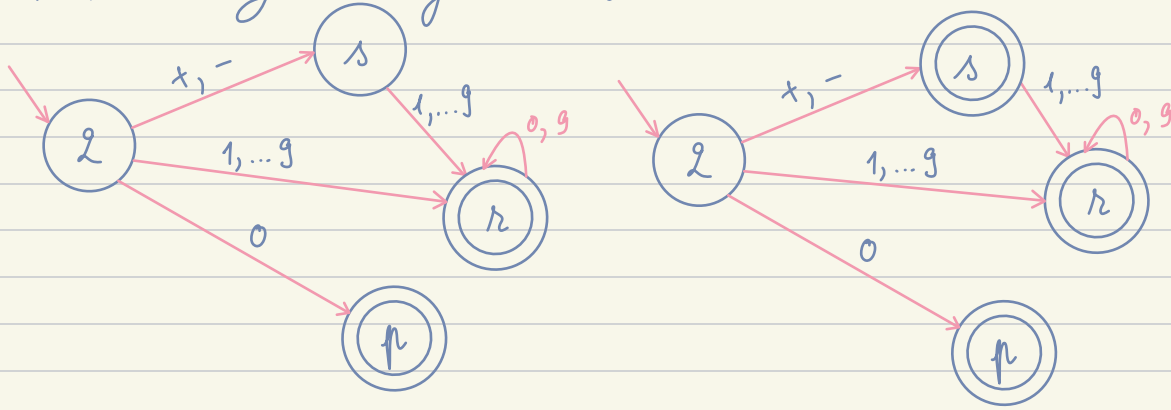
c) $L_3 = \{ c^{3n} \mid n \in \mathbb{N} \}$



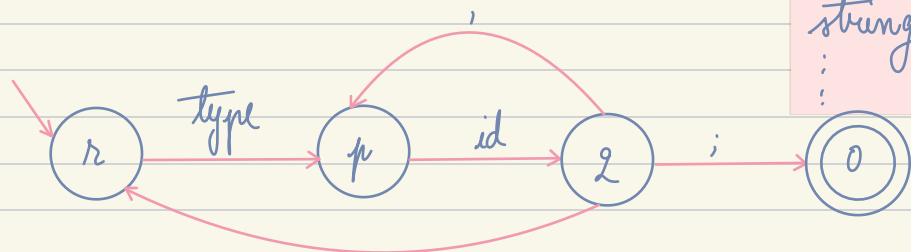
d) $L_4 = \{ x00y \mid x, y \in \{0,1\}^* \}$



e) ? FA for integer numbers



f) ? FA for variable decl.



```
int a, b, ... ;
string c, d, ... ;
... ;
```

Seminar 5.

FA $\Leftrightarrow$ RG $\Leftrightarrow$ RE

I. 1. FA $\Leftrightarrow$ RG

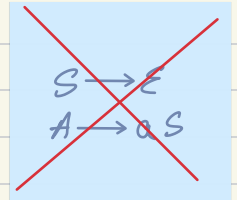
$$G = (\{S, A\}, \{a, b\}, P, S)$$

$$P: S \rightarrow aA$$

$$A \rightarrow aA \mid bA \mid a \mid b$$

? $\Leftrightarrow$ FA

$$RG = \left\{ \begin{array}{l} \text{RLG: } A \rightarrow aB \\ \quad \quad \quad A \rightarrow b \\ \quad \quad \quad \text{if } A \rightarrow \epsilon \in P, A \neq S \\ \quad \quad \quad \text{if } S \in P, \text{ then } S \\ \quad \quad \quad \text{does not appear in rhs} \end{array} \right.$$



$$R = (\{S, A, K\}, \{a, b\}, \dots, \{S\}, \{K\})$$

$$\mathcal{F}: \mathcal{F}(S, a) = A$$

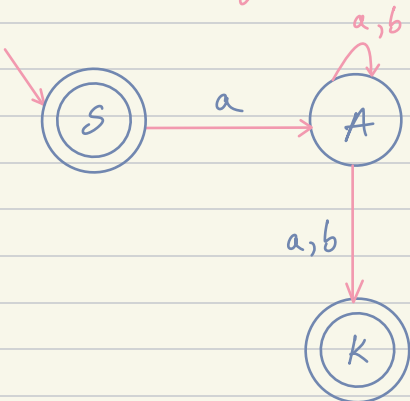
$$\mathcal{F}(A, a) = A$$

throw a we get A

$$\mathcal{F}(A, b) = A$$

$$\mathcal{F}(A, a) = K$$

$$\mathcal{F}(A, b) = K$$



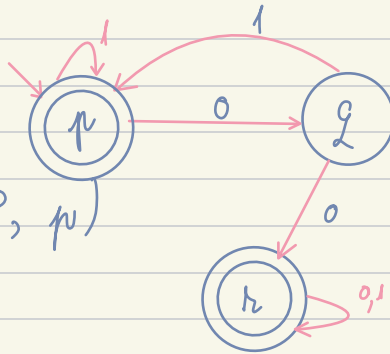
Type	Regular Expression	RLG	LLG
Single terminal	e	$S \rightarrow e$	$S \rightarrow e$
Union operation	$(e + f)$	$S \rightarrow e \mid f$	$S \rightarrow e \mid f$
Concatenation	ef	$S \rightarrow eA, A \rightarrow f$	$S \rightarrow Af, A \rightarrow e$
Star closure	e^*	$S \rightarrow eS \mid \epsilon$	$S \rightarrow Se \mid \epsilon$
Plus closure	e^+	$S \rightarrow eS \mid e$	$S \rightarrow Se \mid e$
Star closure on union	$(e + f)^*$	$S \rightarrow eS \mid fS \mid \epsilon$	$S \rightarrow Se \mid Sf \mid \epsilon$
Plus closure on union	$(e + f)^+$	$S \rightarrow eS \mid fS \mid e \mid f$	$S \rightarrow Se \mid Sf \mid e \mid f$
Star closure on concatenation	$(ef)^*$	$S \rightarrow eA \mid \epsilon;$ $A \rightarrow fS$	$S \rightarrow Af \mid \epsilon;$ $A \rightarrow Se$
Plus closure on concatenation	$(ef)^+$	$S \rightarrow eA ;$ $A \rightarrow fS \mid f$	$S \rightarrow Af ;$ $A \rightarrow Se \mid e$

2. $\neg FA \Rightarrow \neg FLG$
 $? (\Rightarrow) \neg FLG$

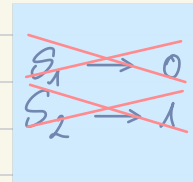
$$G = (\{p, q, r\}, \{0, 1\}, P, p)$$

P:

$p \rightarrow 1p$	$0q$	ϵ	1
$q \rightarrow 1p$	$0r$	0	1
$r \rightarrow 0r$	$1r$	0	1



II. 3. $RG \Leftrightarrow RE$
 $RE: 0(0+1)^* 1$
 $? (\Rightarrow) RG$

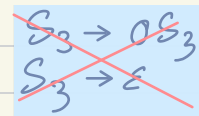


$$\begin{aligned} 0: S_1 &\rightarrow 0 \mid 0S_1 \\ 1: S_2 &\rightarrow 1 \\ 0+1: S_3 &\rightarrow 0 \mid 1 \mid 0S_1 \end{aligned}$$

Kleene star = have any number in any order or none of them

$$\begin{aligned} 0: G_1 &= (\{S_1\}, \{0\}, \{S_1 \rightarrow 0\}, S_1) \\ 1: G_2 &= (\{S_2\}, \{1\}, \{S_2 \rightarrow 1\}, S_2) \end{aligned}$$

$$\begin{aligned} 0+1: G_3 &= (\{S_1, S_2, S_3\}, \{0, 1\}, \{S_1 \rightarrow 0, S_2 \rightarrow 1, S_3 \rightarrow 0 \mid 1\}, S_3) \\ G_3 &= (\{S_3\}, \{0, 1\}, \{S_3 \rightarrow 0 \mid 1\}, S_3) \end{aligned}$$



$$\begin{aligned} 0+1 &\rightarrow \{0, 1\} \\ 0^+ &\rightarrow \{0, 0^2, 0^3, \dots\} \end{aligned} \quad \begin{array}{l} \text{simplified} \\ \text{version} \end{array}$$

$(0+1)^*$:

- $G_4 = (\{S_3\}, \{0, 1\}, \{S_3 \rightarrow \epsilon \mid 0 \mid 1 \mid 0S_3 \mid 1S_3\}, S_3)$
- $G_4 = (\{S_3\}, \{0, 1\}, \{S_3 \rightarrow \epsilon \mid 0S_3 \mid 1S_3\}, S_3)$! not regular

RG: $G_1 \rightarrow G_4$
 $G_2 \rightarrow L_2$
 $G_3 \rightarrow L_1 L_2$ $xy \in L_2$
 $S_3 \Rightarrow (0+1)^+$
 $S \Rightarrow aA \Rightarrow abB \Rightarrow abcc \Rightarrow abcd \quad c \Rightarrow d$

$G_1 \rightarrow L_1$
 $? G_2 \rightarrow L_1^*$

$0(0+1)^*$:

• $G_5 = (\{S_1, S_3\}, \{0, 1\}, \{S_3 \rightarrow 0S_3, S_3 \rightarrow \epsilon \mid 0S_3 \mid 1S_3\}, S_1)$
! not regular

$0(0+1)^* 1$:

• $G_6 = (\{S_1, S_2, S_3\}, \{0, 1\}, \{S_1 \rightarrow 0S_3, S_3 \rightarrow S_2 \mid 0S_3 \mid 1S_3, S_2 \rightarrow 1, S_1\})$! not regular (not RL or LL)
 • $G_6' = (\{S_1, S_3\}, \{0, 1\}, \{S_1 \rightarrow 0S_3, S_3 \rightarrow 1 \mid 0S_3 \mid 1S_3\}, S_1)$

2. RG $\Rightarrow$ RF

$G = (\{S, A, B\}, \{a, b\}, P, S)$

$P: S \rightarrow aA$

$A \rightarrow aA \mid bB \mid b$

$B \rightarrow bB \mid b$

? $\Rightarrow$ RF $a^+ b^+$

$$\begin{cases} S = aA \\ A = aA + bB + b \\ B = bB + b \\ B = b^+ b = b^+ \end{cases}$$

$A = aA + bb^+ + b \Rightarrow A = a^+ b^+$

$S = aA = aa^+ b^+ = a^+ b^+$

$bb^+ \neq b^+$

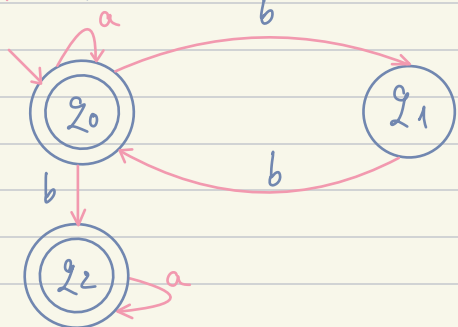
$x = aX + b \Rightarrow X = a^+ b$

$b^+ = bb^*$

$a + ab = a(\epsilon + b)$

III. FA $\Leftrightarrow$ RE

1. FA $\Rightarrow$ RE



$$x = xa + b \quad x = ba^*$$

	a	b
q0	{q0}	{q1, q2}
q1	$\emptyset$	{q0}
q2	{q2}	$\emptyset$

$$\begin{cases} q_0 = \epsilon + q_0 a + q_1 b \\ q_1 = q_0 b \\ q_2 = q_0 b + q_2 a \end{cases}$$

$$bb^+ \neq b^+$$

$$q_0 + q_2 = (a + bb)^* + (a + bb)^* ba^*$$

$$q_0 = \epsilon + q_0 a + q_0 bb$$

$$q_0 = q_0 (a + bb)^* + \epsilon$$

$$q_0 = (a + bb)^*$$

$$q_1 = (a + bb)^* b$$

$$q_2 = (a + bb)^* b + q_2 a$$

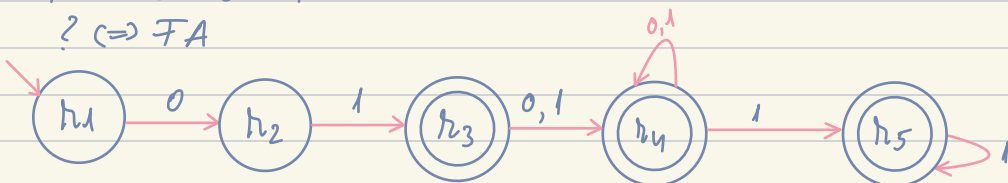
$$q_2 = (a + bb)^* ba^*$$

$$\Rightarrow RE = (a + bb)^* + (a + bb)^* ba^* = (a + bb)^* (\epsilon + ba^*)$$

2. RE $\Rightarrow$ FA

$$RE: 01(0+1)^* 1^*$$

? $\Rightarrow$ FA



Seminar 7.

CFG

1. $G = (\{S, A, B\}, \{0, 1\}, P, S)$

P: $S \rightarrow 0B \mid 1A$

$$A \rightarrow 0 \mid 0S \mid 1AA$$
$$B \rightarrow 1 \mid 1S \mid 0BB$$
$$W = 000110110$$

? build left/right most deriv. for w + parse trees

left most
188668

I. 1886686723

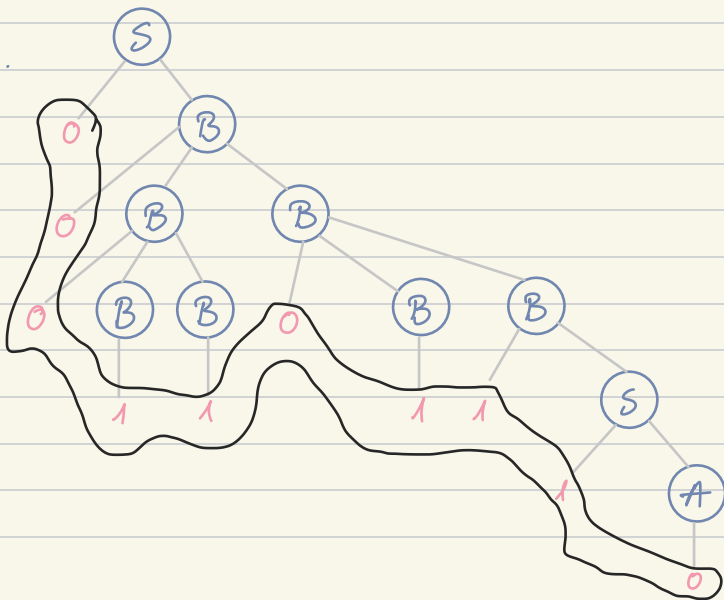
1. $S \Rightarrow 0B \Rightarrow 00BB \Rightarrow 000BBB \Rightarrow 0001BB \Rightarrow 00011B \Rightarrow$

$$\Rightarrow 0001102B \Rightarrow 0001101B \Rightarrow 00011011S \Rightarrow 00011011A \Rightarrow$$

8 6 7 2 5

$\Rightarrow$ 0001101110

三.



right most

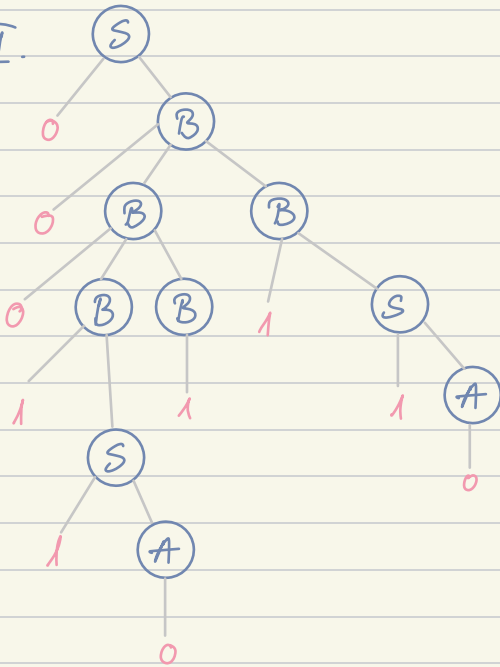
I. 1872386723

II. $S \Rightarrow 0B \Rightarrow 00BB \Rightarrow 00B1S \Rightarrow 00B11A \Rightarrow 00B110 \Rightarrow$
1 8 7 2 3 8

$\Rightarrow 000BB110 \Rightarrow 000B110 \Rightarrow 0001S110 \Rightarrow 00011A110 \Rightarrow$
8 6 7 2 3

$\Rightarrow 0001101110$
3

III.



2. ? G is ambiguous or not

a. $G_1 = (\{S, B, C\}, \{a, b, c\}, P, S)$

$P: S \rightarrow a^1 b^1 c^1 \mid a^2 B$
 $B \rightarrow b^3 C$
 $C \rightarrow c^1$

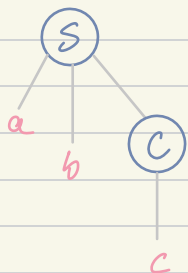
b. $G_2 = (\{E\}, \{a, +, *, (,)\}, P, E)$
 $P: E \rightarrow E + E \mid E * E \mid (E) \mid a$

• $w = abc$

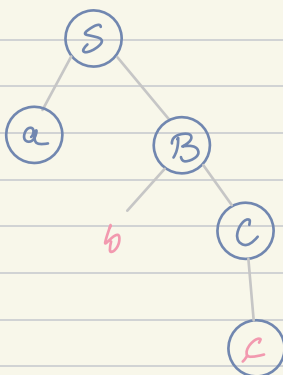
$S \Rightarrow_1 abc \Rightarrow_2 abc$

$S \Rightarrow_2 aB \Rightarrow_3 abc \Rightarrow_4 abc$

tree 1:



tree 2:

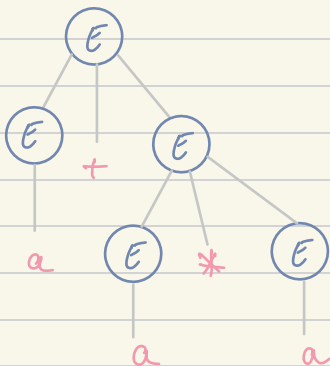


• $w = a + a * a$

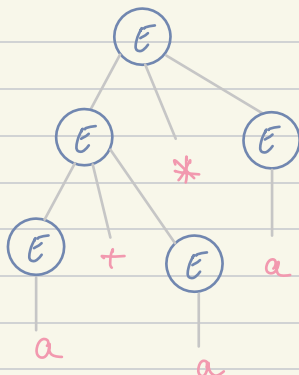
$E \Rightarrow_1 E + E \Rightarrow_4 a + E \Rightarrow_2 a + E * E \Rightarrow_2 a + a * E \Rightarrow_2 a + a * a$

$E \Rightarrow_2 E * E \Rightarrow_1 E + E * E \Rightarrow_4 a + E * E \Rightarrow_4 a + a * E \Rightarrow_4 a + a * a$

tree 1 :



tree 2 :



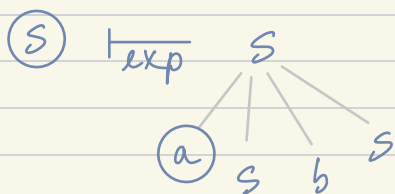
Recursive descent parser

$$G = (\{S\}, \{a, b, c\}, \{S \xrightarrow{S_1} aSbS \mid \xrightarrow{S_2} aS \mid \xrightarrow{S_3} c\}, S)$$

$w = aacbc$

? $w \in L(G)$

$(1, i, \alpha, \beta)$
 $\{q, b, \epsilon, e\}$



$aacbc$

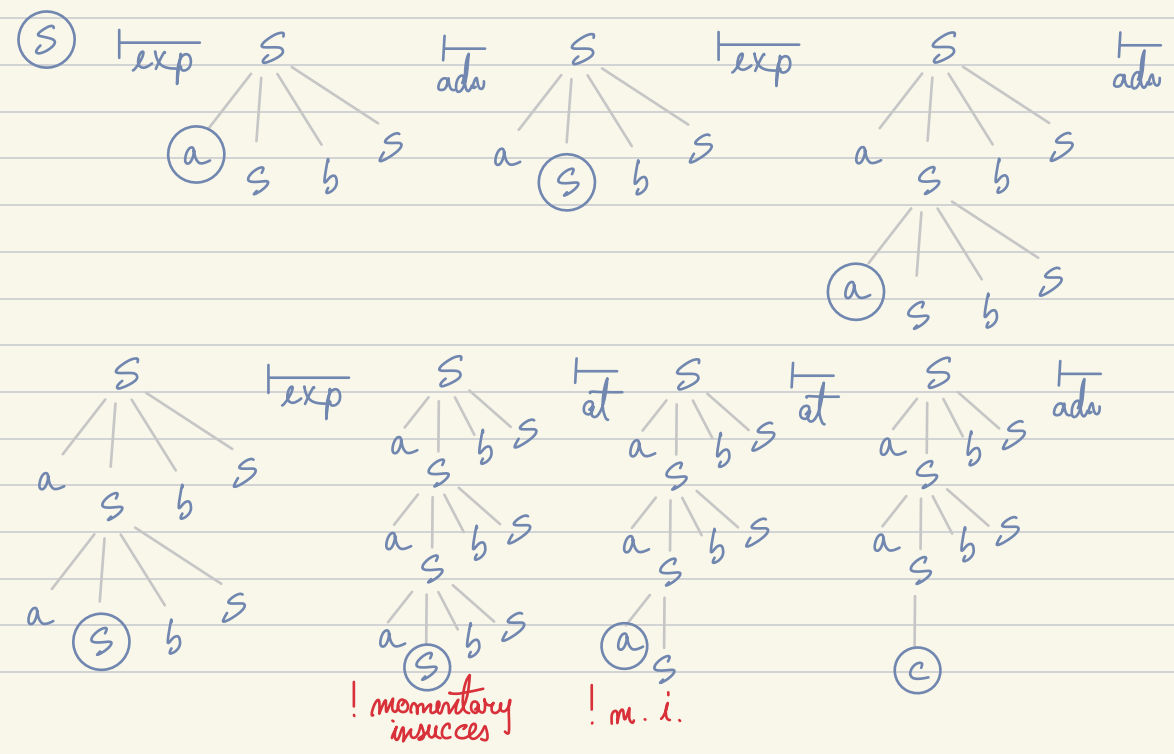
$$(q, 1, \epsilon, S) \xrightarrow{\text{exp}} (q, 1, S_1, aSbS) \xrightarrow{\text{adv}} (q, 2, S_1a, SbS) \xrightarrow{\text{exp}}$$

$$(q, 2, S_1aS_1, aSbSbS) \xrightarrow{\text{adv}} (q, 3, S_1aS_1a, SbSbS) \xrightarrow{\text{exp}}$$

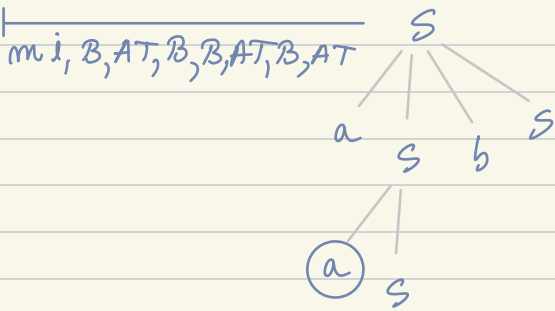
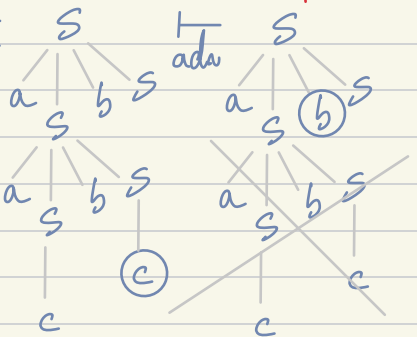
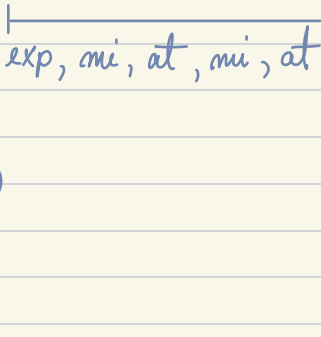
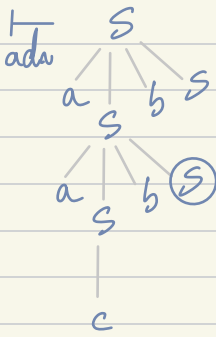
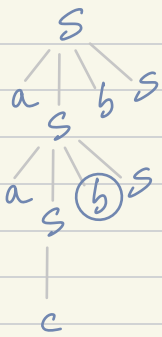
$$(q, 3, S_1aS_1aS_1, aSbSbSbS) \xrightarrow{\text{mi+at}}^2$$

$$(q, 3, S_1aS_1aS_1, aSbSbSbS) \xrightarrow{\text{at}} (q, 3, S_1aS_1aS_2, aSbSbS)$$

$$\begin{aligned}
& \frac{}{\text{mi+at}} (q, 3, S_1 a S_1 a S_3, c b S b S) \frac{}{\text{adv}} (q, 4, S_1 a S_1 a S_3 c, b S b S) \frac{}{\text{adv}} \\
& (q, 5, S_1 a S_1 a S_3 c b, S b S) \frac{}{\text{exp, mi, at, mi, at}} \\
& (q, 5, S_1 a S_1 a S_3 c b S_3, c b S) \frac{}{\text{adv}} (q, 6, S_1 a S_1 a S_3 c b S_3 c, b S) \\
& \frac{}{\text{mi}} (b, 6, S_1 a S_1 a S_3 c b S_3 c, b S) \frac{}{\text{back}} \\
& (b, 5, S_1 a S_1 a S_3 c b S_3, c b S) \frac{}{\text{at}} (b, 5, S_1 a S_1 a S_3 c b, S b S) \\
& \frac{}{\text{back, back, at}} (b, 3, S_1 a S_1 a, S b S b S) \frac{}{\text{back}} (b, 2, S_1 a S_1 a, S b S b S) \\
& \frac{}{\text{at}} (q, 2, S_1 a S_2, a S b S)
\end{aligned}$$



! m.i



Seminar 8 **LL(1) parser**

$G = (\{S, A, B, C, D\}, \{a, +, *, (,)\}, P, S)$

P: (1) $S \rightarrow BA$

(2) $A \rightarrow +BA$

$w = a*(a+a)$

(3) $A \rightarrow \epsilon$

(4) $B \rightarrow DC$

(5) $C \rightarrow *DC$

(6) $C \rightarrow \epsilon$

(7) $D \rightarrow (S)$

(8) $D \rightarrow a$

I. FIRST & FOLLOW

$$\underbrace{\text{FIRST}(X_1 X_2 \dots X_m)}_{\neq} = \text{FIRST}(X_1) \oplus \text{FIRST}(X_2) \oplus \dots \oplus \text{FIRST}(X_m)$$

$\forall i = 1, m, x_i \in N \cup \Sigma$

	F_0	F_1	F_2	F_3	F_4
S	$\emptyset$	$\emptyset$	$\{(, a)\}$	$\{(, a)\}$	
A	$\{+, \epsilon\}$	$\{+, \epsilon\}$	$\{+, \epsilon\}$	$\{+, \epsilon\}$	
B	$\emptyset$	$\{(, a)\}$	$\{(, a)\}$	$\{(, a)\}$	
C	$\{*, \epsilon\}$	$\{*, \epsilon\}$	$\{*, \epsilon\}$	$\{*, \epsilon\}$	
D	$\{(, a)\}$	$\{(, a)\}$	$\{(, a)\}$	$\{(, a)\}$	

FIRST

$S \xRightarrow{*} S$

$\text{FIRST}(S) = \{(, a)\}$
 $\text{FIRST}(A) = \{+, \epsilon\}$
 $\text{FIRST}(B) = \{(, a)\}$
 $\text{FIRST}(C) = \{*, \epsilon\}$
 $\text{FIRST}(D) = \{(, a)\}$

III. Parse the seq. w

$(a * (a + a) \$, S \$ \epsilon) \vdash (a * (a + a) \$, BA \$, 1) \vdash$
 $(a * (a + a) \$, \Delta CA \$, 14) \vdash (a * (a + a) \$, a CA \$, 148) \vdash$
 $(* (a + a) \$, CA \$, 148) \vdash (* (a + a) \$, * \Delta CA \$, 1485) \vdash$
 $((a + a) \$, \Delta CA \$, 1485) \vdash ((a + a) \$, (S) CA \$, 14857) \vdash$
 $(a + a) \$ S) CA \$, 14857) \vdash (a + a) \$, BA) CA \$, 148571) \vdash$
 $(a + a) \$, \Delta CA) CA \$, 148714) \quad (a + a) \$, a CA) CA \$, 14857148) \vdash$
 $(+ a) \$, CA) CA \$, 14857148) \vdash (+ a) \$, A) CA \$, 148571486) \vdash$
 $(+ a) \$, + BA) CA \$, 1485714862) \vdash (a) \$, BA) CA \$, 1485714862) \vdash$
 $(a) \$, \Delta CA) CA \$, 14857148624) \vdash (a) \$, a CA) CA \$, 148571486248) \vdash$
 $() \$, CA) A \$, 148571486248) \vdash \underline{(\$ \$, 1485714862486363)}$

Conflict

$$\begin{array}{l} A \rightarrow \underline{\alpha} \beta \\ A \rightarrow \underline{\alpha} \gamma \end{array} \Rightarrow \left\{ \begin{array}{l} A \rightarrow \alpha \beta \\ B \rightarrow \beta / \gamma \end{array} \right.$$

Both options are correct, but because first one has productions starting both with α the LL(1) does not know which one to use.

Seminar 9.

LR(0) parser

$$G = (\{S, A\}, \{a, b, c\}, P, S)$$

$$S' \rightarrow S$$

$$P: \begin{aligned} (1) & S \rightarrow a A \\ (2) & A \rightarrow b A \\ (3) & A \rightarrow c \end{aligned}$$

$$w = a b b c$$

$$? w \in L(G)$$

I Canonical collection of states (set)

LR(k) item	$[A \rightarrow \alpha \cdot \beta, \mu]$
	$\swarrow$ kernel $\searrow$ prod
LR(0) item	$[A \rightarrow \alpha \cdot \beta] \sum^k$

$$\begin{pmatrix} [A \rightarrow \alpha \cdot B \beta] \\ [B \rightarrow \cdot \gamma] \end{pmatrix} \quad \text{EN}$$

$$\mathcal{C} = \{s_0, s_1, \dots\}$$

$$s_0 = \text{closure}(\{[S' \rightarrow \cdot S]\}) = \{[S' \rightarrow \cdot S], [S \rightarrow \cdot a A]\}$$

$$s_1 = \text{goto}(s_0, S) = \text{closure}(\{[S \rightarrow S \cdot]\}) = \{[S \rightarrow S \cdot]\}$$

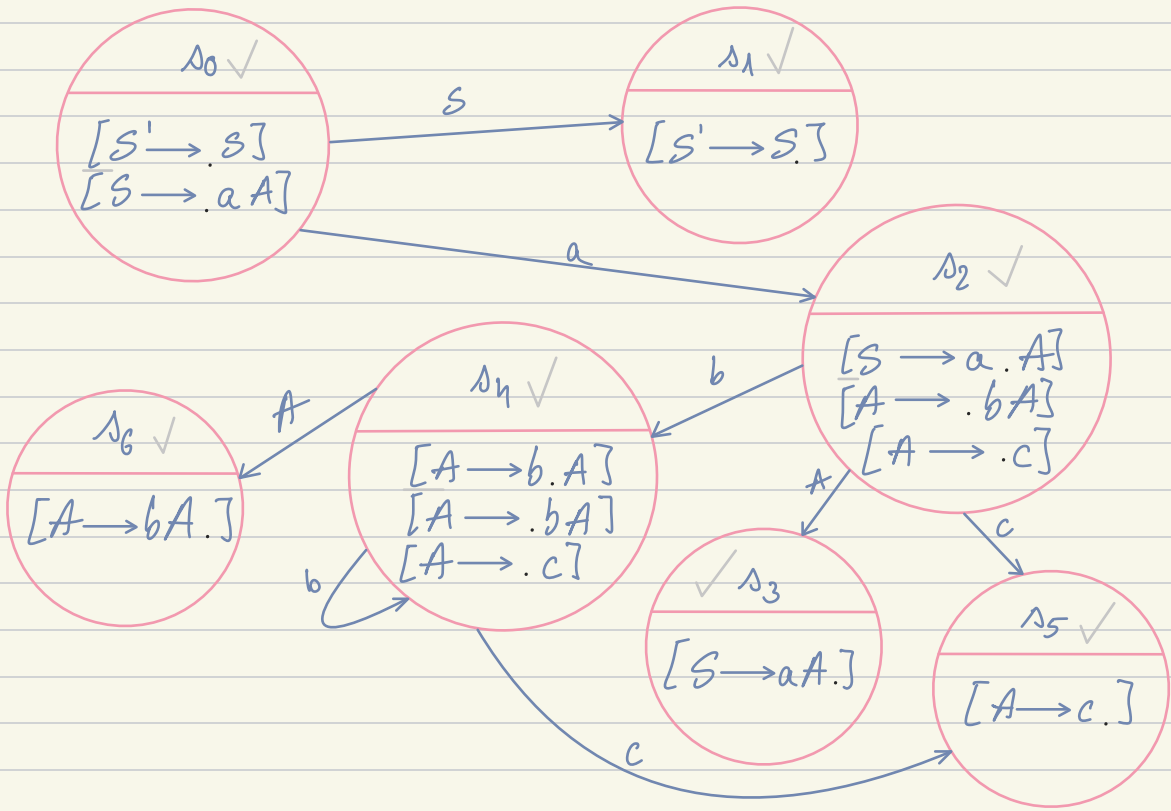
$$\text{goto}(s_0, A) = \emptyset$$

$$s_2 = \text{goto}(s_0, a) = \text{closure}(\{[S \rightarrow a \cdot A]\}) = \{[S \rightarrow a \cdot A], [A \rightarrow \cdot b A], [A \rightarrow \cdot c]\}$$

$$\text{goto}(s_0, b) = \text{goto}(s_0, c) = \emptyset$$

$$s_3 = \text{goto}(s_2, A) = \text{closure}(\{[S \rightarrow a A \cdot]\}) = \{[S \rightarrow a A \cdot]\}$$

$$s_4 = \text{goto}(s_2, b) = \text{closure}(\{[A \rightarrow b \cdot A]\}) = \{[A \rightarrow b \cdot A], [A \rightarrow \cdot b A], [A \rightarrow \cdot c]\}$$



$$\begin{aligned}
 \Lambda_5 &= \text{goto}(\Lambda_2, c) = \text{closure}(\{[A \rightarrow c \cdot]\}) = \{[A \rightarrow c \cdot]\} \\
 \Lambda_6 &= \text{goto}(\Lambda_4, A) = \text{closure}(\{[A \rightarrow bA \cdot]\}) = \{[A \rightarrow bA \cdot]\} \\
 \text{goto}(\Lambda_4, b) &= \text{closure}(\{[A \rightarrow b \cdot A]\}) = \Lambda_4 \\
 \text{goto}(\Lambda_4, c) &= \text{closure}(\{[A \rightarrow \cdot c]\}) = \Lambda_5 \\
 \text{goto}(\Lambda_0, A) &= \emptyset \\
 \Lambda_2 &= \text{goto}(\Lambda_0, a) = \text{closure}(\{[S \rightarrow a \cdot A], [A \rightarrow \cdot bA], [A \rightarrow \cdot c]\}) \\
 \text{goto}(\Lambda_0, b) &= \text{goto}(\Lambda_0, c) = \emptyset
 \end{aligned}$$

II. LR(0) parsing table

	action	goto				
		S	A	a	b	C
Δ_0	shift	Δ_1		Δ_2		
Δ_1	accept					
Δ_2	shift		Δ_3		Δ_4	Δ_5
Δ_3	reduced 1					
Δ_4	shift		Δ_6		Δ_4	Δ_5
Δ_5	reduced 3					
Δ_6	reduced 2					

III. Parse in

work	input	output
$\$ \Delta_0$	a.bbc\$	ϵ
$\$ \Delta_0 a \Delta_2$	bbc\$	ϵ
$\$ \Delta_0 a \Delta_2 b \Delta_4$	bc\$	ϵ
$\$ \Delta_0 a \Delta_2 b \Delta_4 b \Delta_4$	c\$	ϵ
$\$ \Delta_0 a \Delta_2 b \Delta_4 b \Delta_4 c \Delta_5$	\$	ϵ
$\$ \Delta_0 a \Delta_2 b \Delta_4 b \Delta_4 \Delta_6$	\$	3
$\$ \Delta_0 a \Delta_2 b \Delta_4 \Delta_6$	\$	23
$\$ \Delta_0 a \Delta_2 \Delta_3$	\$	223
$\$ \Delta_0 \Delta_1$	\$	1223
\$accept		

Seminar 11

SLR

1. $G = (\{S', E, T\}, \{+, id, const, (,)\}, P, S)$

$$P: S \rightarrow S$$

$$(1) E \rightarrow T$$

$$(2) E \rightarrow E + T$$

$$(3) T \rightarrow (E)$$

$$(4) T \rightarrow id$$

$$(5) T \rightarrow const$$

$$w = id + const$$

$$[A \rightarrow \alpha \beta]$$

$$\bar{I} \quad \Delta_0 = \text{closure}(\{[S' \rightarrow \cdot E]\}) = \{[S' \rightarrow \cdot E], [E \rightarrow \cdot T], [E \rightarrow \cdot E + T], [T \rightarrow \cdot (E)], [T \rightarrow \cdot id], [T \rightarrow \cdot const]\}$$

$$\Delta_1 = \text{goto}(\Delta_0, E) = \text{closure}(\{[S' \rightarrow E \cdot], [E \rightarrow E \cdot + T]\}) = \{[S' \rightarrow E \cdot], [E \rightarrow E \cdot + T]\}$$

$$\Delta_2 = \text{goto}(\Delta_0, T) = \text{closure}(\{[E \rightarrow T \cdot]\}) = \{[E \rightarrow T \cdot]\}$$

$$\Delta_3 = \text{goto}(\Delta_0, '(') = \text{closure}(\{[T \rightarrow (\cdot E)]\}) = \{[T \rightarrow (\cdot E)], [E \rightarrow \cdot T], [E \rightarrow \cdot E + T], [T \rightarrow \cdot (E)], [T \rightarrow \cdot id], [T \rightarrow \cdot const]\}$$

$$\Delta_4 = \text{goto}(\Delta_0, id) = \text{closure}(\{[T \rightarrow id \cdot]\}) = \{[T \rightarrow id \cdot]\}$$

$$\Delta_5 = \text{goto}(\Delta_0, const) = \text{closure}(\{[T \rightarrow const \cdot]\}) = \{[T \rightarrow const \cdot]\}$$

$$\Delta_6 = \text{goto}(\Delta_1, +) = \text{closure}(\{[E \rightarrow E + \cdot T]\}) = \{[E \rightarrow E + \cdot T], [T \rightarrow \cdot (E)], [T \rightarrow \cdot id], [T \rightarrow \cdot const]\}$$

$$\Delta_7 = \text{goto}(\Delta_3, E) = \text{closure}(\{[T \rightarrow (E \cdot)], [E \rightarrow E \cdot + T]\}) = \{[T \rightarrow (E \cdot)], [E \rightarrow E \cdot + T]\}$$

$$\text{goto}(\Delta_3, T) = \text{closure}(\{[E \rightarrow T \cdot]\}) = \Delta_2 \quad \text{goto}(\Delta_3, '(') = \Delta_3$$

$$\text{goto}(\Delta_3, id) = \Delta_4 \quad \text{goto}(\Delta_3, const) = \Delta_5 \quad \text{goto}(\Delta_3, +) = \Delta_6$$

$$\Delta_8 = \text{goto}(\Delta_6, T) = \text{closure}(\{[E \rightarrow E + T \cdot]\}) = \{[E \rightarrow E + T \cdot]\}$$

$$\text{goto}(\Delta_6, '(') = \Delta_3 \quad \text{goto}(\Delta_6, id) = \Delta_4 \quad \text{goto}(\Delta_6, const) = \Delta_5$$

$$\Delta_9 = \text{goto}(\Delta_7, ')') = \text{closure}(\{[T \rightarrow (E).]\}) = \{[T \rightarrow (E).]\}$$

$$\text{goto}(\Delta_7, +) = \Delta_6$$

$$\text{II. FOLLOW}(E) = \{\epsilon, +,)\} = \text{FOLLOW}(T)$$

	+	(	)	id	const	\$	E	T
Δ_0		shift Δ_3		shift Δ_4	shift Δ_5		Δ_1	Δ_2
Δ_1	shift Δ_6					accept		
Δ_2	reduce 1		reduce 1			reduce 1		
Δ_3		shift Δ_3		shift Δ_4	shift Δ_5		Δ_7	Δ_2
Δ_4	reduce 4		reduce 4			reduce 4		
Δ_5	reduce 5		reduce 5			reduce 5		
Δ_6		shift Δ_3		shift Δ_4	shift Δ_5			Δ_8
Δ_7	shift Δ_6		shift Δ_3					
Δ_8	reduce 2		reduce 2			reduce 2		
Δ_9	reduce 3		reduce 3			reduce 3		

III.	work	input	output
	\$ Δ_0	id + const \$	ϵ
	\$ Δ_0 id Δ_4	+ const \$	ϵ
	\$ Δ_0 T Δ_2	+ const \$	4
	\$ Δ_0 E Δ_1	+ const \$	14
	\$ Δ_0 E Δ_1 + Δ_6	const \$	14
	\$ Δ_0 E Δ_1 + Δ_6 const Δ_5	\$	14
	\$ Δ_0 E Δ_1 + Δ_6 T Δ_8	\$	514
	\$ Δ_0 E Δ_1	\$	2514
	\$ accept		

Seminar 12 **LR(1)**

1. $G = (\{S', S, A\}, \{a, b\}, P, S')$

$P: S' \rightarrow S$

(1) $S' \rightarrow AA$

(2) $A \rightarrow aA$

(3) $A \rightarrow b$

$w = abab$

LR(1) item $[A \rightarrow \alpha \beta, a]$

$\swarrow$
 $e \in \Sigma \cup \{\$ \}$

I Canonical collection

$FIRST(A) = \{a, b\} = FIRST(S)$
 $[A \rightarrow \alpha \beta, a] \rightarrow [B \rightarrow \gamma, b]$
 $[S' \rightarrow \cdot S, \$]$

$\uparrow FIRST(\beta a)$

$S \rightarrow AB$
 $A \rightarrow aA \mid b \mid \epsilon$
 $B \rightarrow c$

$s_0 = \text{closure}(\{[S' \rightarrow \cdot S, \$]\}) = (\{[S' \rightarrow \cdot S, \$], [S \rightarrow \cdot AA, \$], [A \rightarrow \cdot aA, a], [A \rightarrow \cdot aA, b], [A \rightarrow \cdot b, a], [A \rightarrow \cdot b, b]\})$
 $s_1 = \text{goto}(s_0, S) = \text{closure}(\{[S' \rightarrow S, \$]\}) = \{[S' \rightarrow S, \$]\}$
 $s_2 = \text{goto}(s_0, A) = \text{closure}(\{[S \rightarrow A \cdot A, \$]\}) = \{[S \rightarrow A \cdot A, \$], [A \rightarrow \cdot aA, \$], [A \rightarrow \cdot b, \$]\}$
 $s_3 = \text{goto}(s_0, a) = \text{closure}(\{[A \rightarrow a \cdot A, a], [A \rightarrow a \cdot A, b]\}) = \{[A \rightarrow a \cdot A, a], [A \rightarrow a \cdot A, b], [A \rightarrow \cdot aA, a], [A \rightarrow \cdot aA, b], [A \rightarrow \cdot b, a], [A \rightarrow \cdot b, b]\}$
 $s_4 = \text{goto}(s_0, b) = \text{closure}(\{[A \rightarrow b \cdot, a], [A \rightarrow b \cdot, b]\}) = \{[A \rightarrow b \cdot, a], [A \rightarrow b \cdot, b]\}$
 $s_5 = \text{goto}(s_2, A) = \text{closure}(\{[S \rightarrow AA \cdot, \$]\}) = \{[S \rightarrow AA \cdot, \$]\}$
 $s_6 = \text{goto}(s_2, a) = \text{closure}(\{[A \rightarrow a \cdot A, \$]\}) = \{[A \rightarrow a \cdot A, \$], [A \rightarrow \cdot aA, \$], [A \rightarrow \cdot b, \$]\}$

$$\begin{aligned}
 \Delta_7 &= \text{goto}(\Delta_2, b) = \text{closure}(\{[A \rightarrow b., \$]\}) = \{[A \rightarrow b., \$]\} \\
 \Delta_8 &= \text{goto}(\Delta_3, A) = \text{closure}(\{[A \rightarrow aA., a], [A \rightarrow aA., b]\}) = \\
 &= \{[A \rightarrow aA., a], [A \rightarrow aA., b]\} \\
 \Delta_9 &= \text{goto}(\Delta_3, a) = \dots = \Delta_3 \quad \text{goto}(\Delta_3, b) = \Delta_4 \\
 \Delta_6 &= \text{goto}(\Delta_6, A) = \text{closure}(\{[A \rightarrow aA., \$]\}) = \{[A \rightarrow aA., \$]\} \\
 \Delta_6 &= \text{goto}(\Delta_6, a) = \Delta_6 \quad \text{goto}(\Delta_6, b) = \Delta_7
 \end{aligned}$$

II. LR(1) parsing table

	action			goto	
	a	b	\$	S	A
Δ_0	SH Δ_3	SH Δ_4		Δ_1	Δ_2
Δ_1			ACC		
Δ_2	SH Δ_6	SH Δ_7			Δ_5
Δ_3	SH Δ_3	SH Δ_4			Δ_8
Δ_4	R(3)	R(3)			
Δ_5			R(1)		
Δ_6	SH Δ_6	SH Δ_7			Δ_9
Δ_7			R(3)		
Δ_8	R(2)	R(2)			
Δ_9			R(2)		

III. Parse w

work	input	output
\$s0	abab\$	ϵ
\$s0a s3	bab\$	ϵ
\$s0a s3b s4	ab\$	ϵ
\$s0a s3A s8	ab\$	3

work	input	output
\$s0A s2	a b \$	23
\$s0A s2a s6	b \$	23
\$s0A s2a s6b s7	\$	23
\$accept		

LALR(1)

I. Canonical collection

$$C = \{ s_0, s_1, s_2, s_3, s_4, s_5, s_8 \}$$

$$s_3 = \{ [A \rightarrow a.A, a/b/\$], [A \rightarrow .aA, a/b/\$], [A \rightarrow .b, a/b/\$] \}$$

$$s_4 = \{ [A \rightarrow b., a/b/\$] \}$$

$$s_8 = \{ [A \rightarrow aA., a/b/\$] \}$$

II. Parsing table

	action			goto	
	a	b	\$	S	A
s0	SH s3	SH s4		s1	s2
s1			ACC		
s2	SH s3	SH s4			s5
s3	SH s3	SH s4			s8
s4	R(3)	R(3)	R(3)		
s5			R(1)		
s8	R(2)	R(2)	R(2)		

Seminar 13

PDA

if and only if

$$\begin{aligned} (p, \underline{a}w, Z_\gamma) \vdash (q, w, \alpha_\gamma) & \text{ iff } (q, \alpha) \in \delta(p, \underline{a}, Z) \\ (p, w, Z_\gamma) \vdash (q, \tilde{w}, \alpha_\gamma) & \text{ iff } (q, \alpha) \in \delta(p, \underline{\epsilon}, Z) \end{aligned}$$

1. $L_1 = \{ 0^m 1^{2m} \mid m \in \mathbb{N}^* \}$

$$0, x \rightarrow xxx$$

$$0, 2_0 \rightarrow xx2_0$$

$$1, x \rightarrow \epsilon$$

$$w_1 = 0^2 1^4 \in L(M)$$

$$w_2 = 0^2 1^3 \notin L(M)$$

$$w_3 = 0^2 1^5 \notin L(M)$$



$$\begin{aligned} w_1: (q_0, 0^2 1^4, 2_0) & \vdash (q_0, 0^1 1^4, xxx2_0) \vdash (q_0, 1^4, xxxxx2_0) \\ & \vdash (q_1, 1^3, xxx2_0) \vdash (q_1, \epsilon, 2_0) \vdash (q_2, \epsilon, 2_0) \Rightarrow \\ & \Rightarrow w_1 \in L(M) \end{aligned}$$

$$\begin{aligned} w_2: (q_0, 0^2 1^3, 2_0) & \vdash^2 (q_0, 1^3, x^4 2_0) \vdash^3 (q_0, \epsilon, x 2_0) \Rightarrow \\ & \Rightarrow w_2 \notin L(M) \end{aligned}$$

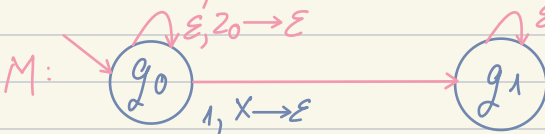
$$\begin{aligned} w_3: (q_0, 0^2 1^5, 2_0) & \vdash^2 (q_0, 1^5, x^4 2_0) \vdash^4 (q_1, 1, 2_0) \vdash \\ & \vdash (q_2, 1, 2_0) \Rightarrow w_3 \notin L(M) \end{aligned}$$

$$0, x \rightarrow xxx$$

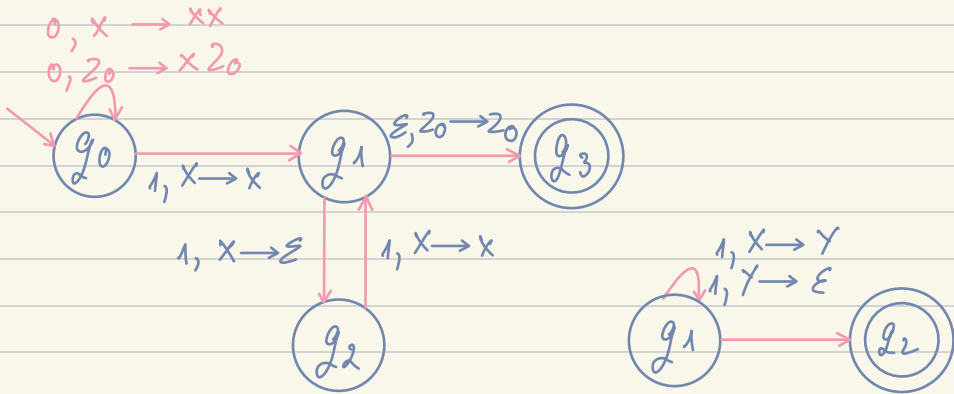
$$0, 2_0 \rightarrow xx2_0$$

$$1, x \rightarrow \epsilon$$

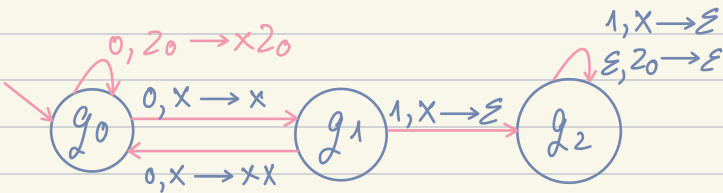
$$\epsilon, 2_0 \rightarrow \epsilon$$



2. I. $L_2 = \{ 0^m (1^2)^m \mid m \in \mathbb{N}^* \}$



II. $L_2 = \{ 0^{2m} 1^m \mid m \in \mathbb{N}^* \}$

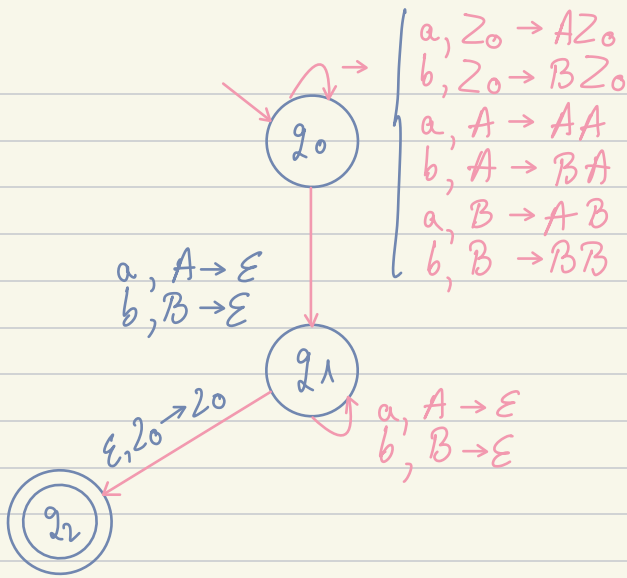


$$\begin{aligned}
 (q_0, 0^4 1^2, 2_0) &\vdash (q_0, 0^3 1^2, x 2_0) \vdash (q_1, 0^2 1^2, x 2_0) \vdash \\
 (q_0, 0 1^2, x x 2_0) &\vdash (q_1, 1^2, x x 2_0) \vdash^2 (q_2, \epsilon, 2_0) \vdash (q_2, \epsilon, \epsilon) \Rightarrow \\
 &\Rightarrow \in L(M)
 \end{aligned}$$

3. $L_3 = \{ \underbrace{w}_{q_0} \underbrace{w}_{q_1}^R \mid w \in \{a, b\}^* \}$

abaa ba
abbbba

A aa
 B
 A
 $?$



Seminar 14.

Attribute grammars

$(G, A, \mathcal{F})$

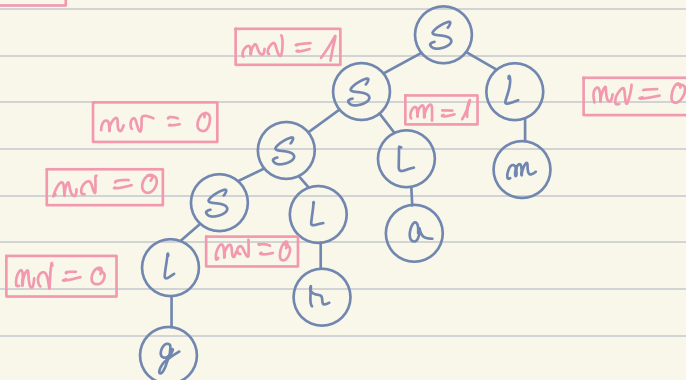
cfg

Given AG for

1. computing mn of vowels in a non-empty string
2. computing the value of an attribute expr. with $+, *, /, -, (,)$
3. checking if $m \geq 3, m \in \mathbb{N}$

1. $S \rightarrow L$ $S.mn = L.mn$
 - $S \rightarrow SL$ $S_1.mn = L_1.mn + S_2.mn$
 - $L \rightarrow a | e | i | o | u | A | E | I | O | U$ $L.mn = 1$
 - $L \rightarrow b | c | \dots | z | B | C | \dots | Z$ $L.mn = 0$
-
- mn

$w = \text{groom}$

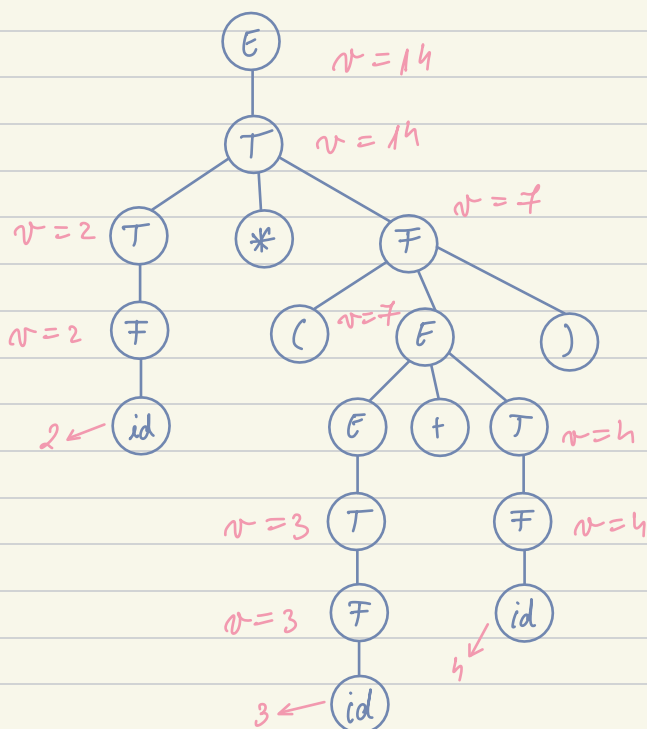


2. $E \rightarrow E + T$	$E_1.v = E_2.v + T.v$
$E \rightarrow E - T$	$E_1.v = E_2.v - T.v$
$E \rightarrow T$	$E.v = T.v$
$T \rightarrow T * F$	$T_1.v = T_2.v * F.v$
$T \rightarrow T / F$	$T_1.v = T_2.v / F.v$
$T \rightarrow F$	$T.v = F.v$
$F \rightarrow (E)$	$F.v = E.v$
$F \rightarrow id$	$F.v = id.v$
$F \rightarrow const$	$F.v = const.v$

v

$w = a * (b + c)$

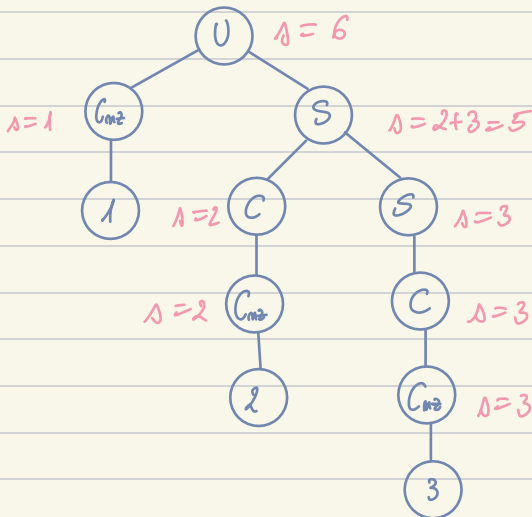
$id * (id + id)$



3	$U \rightarrow C$	$U.\Delta = C.\Delta$
	$U \rightarrow C_{mz} S$	$U.\Delta = C_{mz}.\Delta + S.\Delta$
	$S \rightarrow C$	$S.\Delta = C.\Delta$
	$S \rightarrow CS$	$S.\Delta = S.\Delta + C.\Delta$
	$C_{mz} \rightarrow 1$	$C_{mz}.\Delta = 1$
	...	...
	$C_{mz} \rightarrow 9$	$C_{mz}.\Delta = 9$
	$C \rightarrow 0$	$C.\Delta = 0$
	$C \rightarrow C_{mz}$	$C.\Delta = C_{mz}.\Delta$

$U.isdiv = (U.\Delta \% 3 == 0)$
 $U.isdiv = ((C_{mz}.\Delta + S.\Delta) \% 3 == 0)$

for 123



$\Delta=6 \Rightarrow isdiv = true$

3. Address Code

```

if (a < b) or c AND (e > d)
    then a := -1;
    else a := b + c * 3;
endif

```

index	op	arg 1	arg 2	result
1	<	a	b	t ₁
2	>	b	d	t ₂
3	and	c	t ₂	t ₃
4	or	t ₁	t ₃	t ₄
5	goto	t ₄		(10)
6	*	c	3	t ₅
7	+	b	t ₅	t ₆
8	:=	t ₆		a
9	goto			(12)
10	@	1		t ₇
11	:=	t ₇		a
12	...			

<op> <arg1> <arg2> <result>

```

while (a < b) do
    a := a + 1;
    ...
endwhile

```

index	op	arg 1	arg 2	result
1	<	a	b	
2	>	b	d	
3	and	c	(2)	
4	or	(1)	(3)	
5	goto	(4)	(10)	
6	*	c	3	
7	+	b	(8)	
8	:=	(7)	a	
9	goto		(12)	
10	@	1		
11	:=	(10)	a	
12				

index	op	arg 1	arg 2	result
1	<	a	b	t ₁
2	!	t ₁		t ₂
3	goto	t ₂		(7)
4	+	a		t ₄
5	:=	t ₄	1	a
6	goto			(1)
7				